

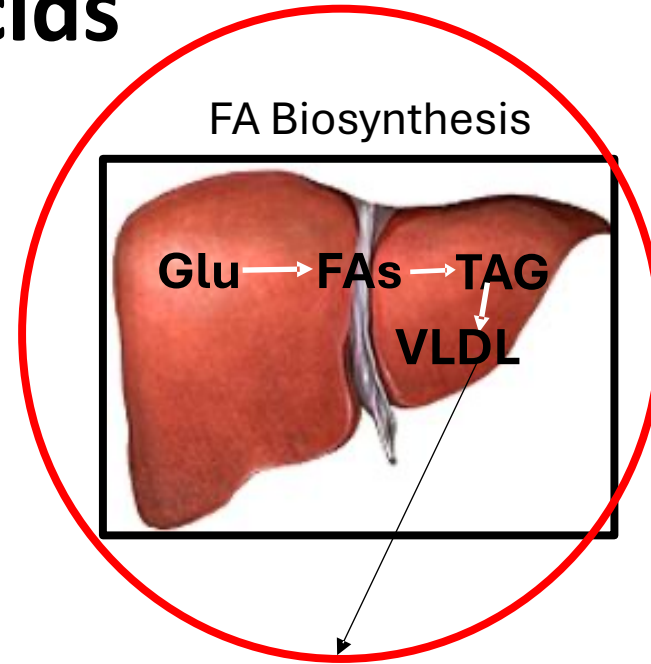
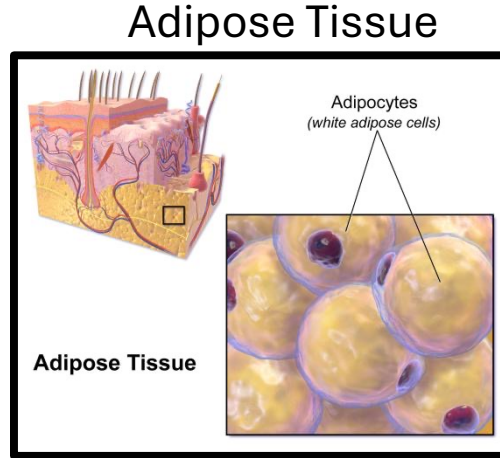
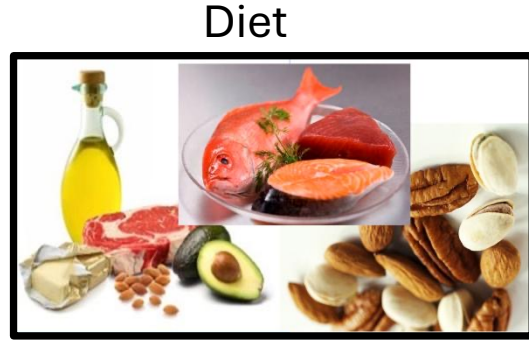
Basic Principles of Medicine 2
Module : Digestion and Metabolism
Lecture No: 16

Fatty Acid Synthesis and Regulation

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Sources of Fatty Acids



TAG

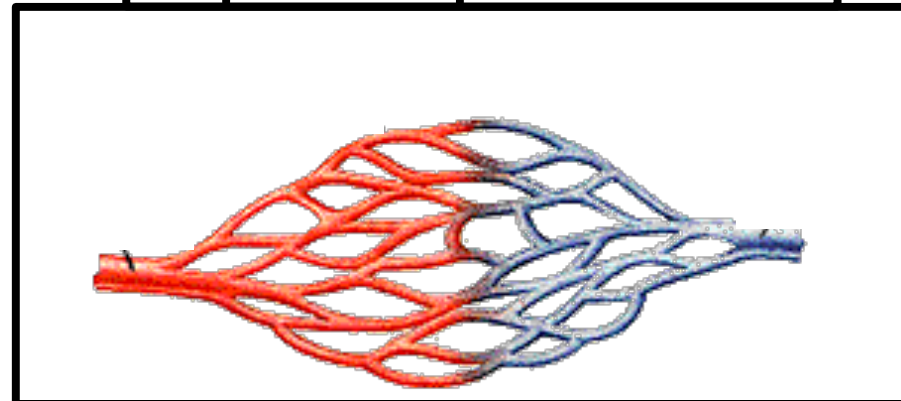
FFA

VLDL

Chylomicrons

Lipid transport in blood

Fatty acids play a central role in metabolism as building blocks of biological membranes, energy reserves, and precursors to second messenger molecules.



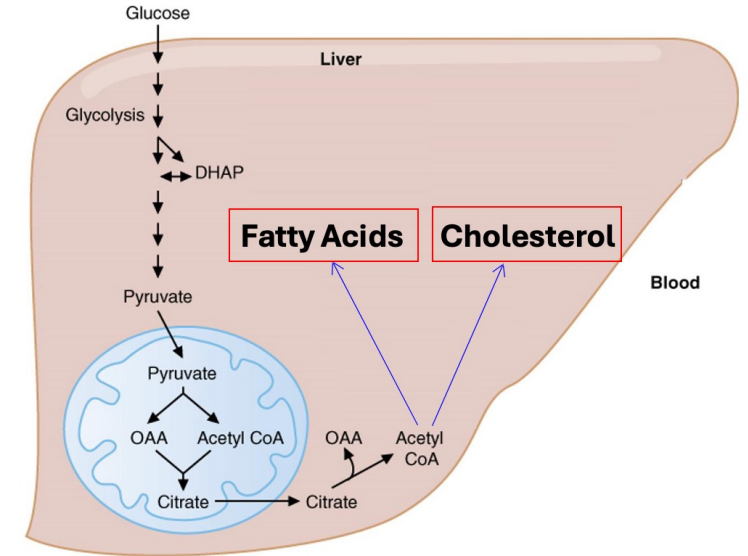
Fatty acids are obtained in liver by hepatocellular uptake from the plasma and by de novo biosynthesis.

Fatty Acid Synthesis

- FA Synthesis *is linked to* Carbohydrate Metabolism

Sites FA Synthesis

- Major sites = Liver + Lactating Mammary gland
- Minor site = Adipose tissue
- Dysregulation of FA synthesis is linked to diseases
(e.g. cancer, obesity, and non-alcoholic fatty liver disease)



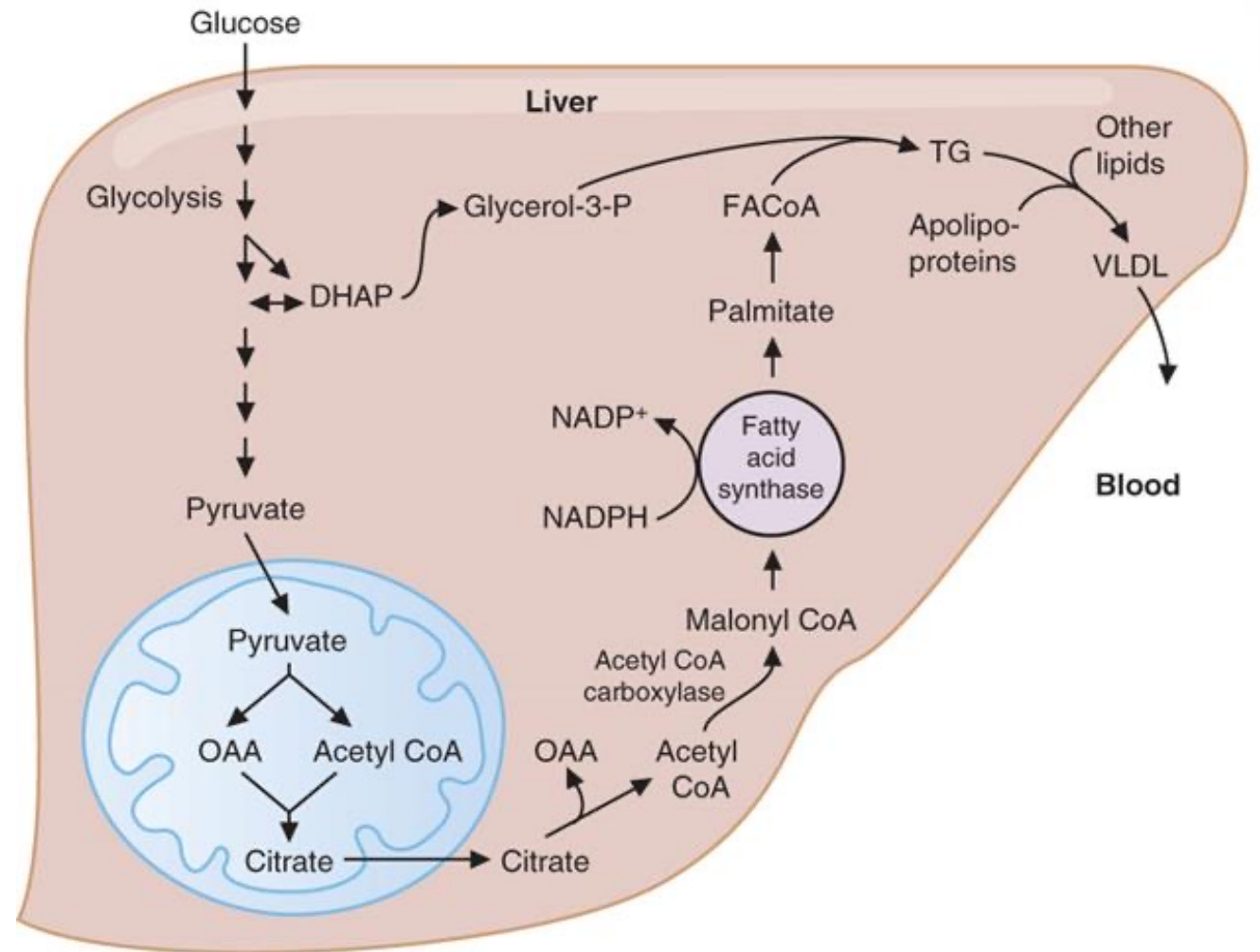
Insulin activates Liver FA & Cholesterol Synthesis

Liver Synthesis of FA

- **FA Synthesis** occurs in **Hepatocyte Cytosol**
- Occurs in the **well-fed state** = **Insulin @ Liver**.

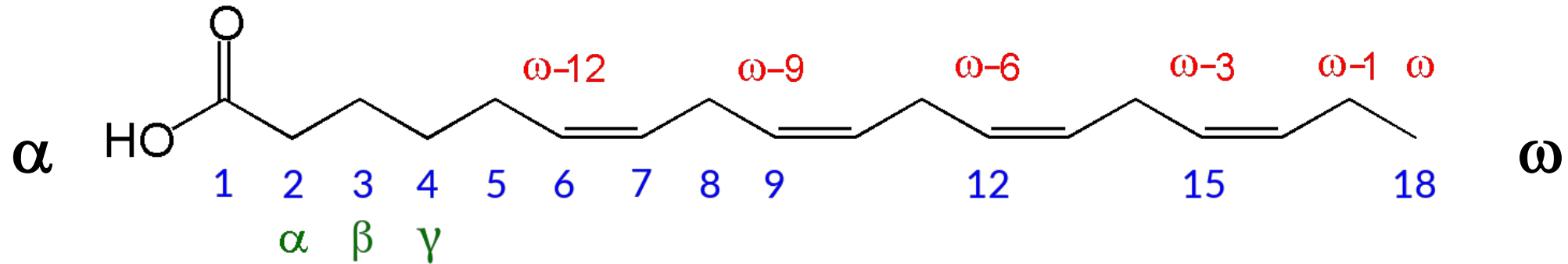
Precursors and Co-enzymes

- **Acetyl CoA**
- **CoA (pantothenic acid)**
- **Biotin**
- **ATP**
- **NADPH**
- **CO₂**

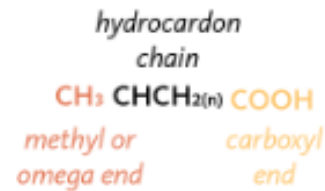


Fatty acid nomenclature

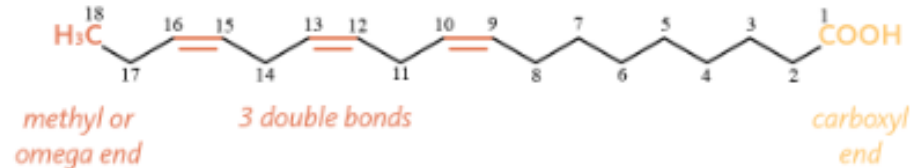
Fatty acid carbon atoms are **numbered** (or lettered with Greek lettering system (α , β , γ , δ) either **from** the **carboxyl** group **OR** from the carbon **furthest** removed from the carboxyl group (ω numbering system)



(a)



(b)

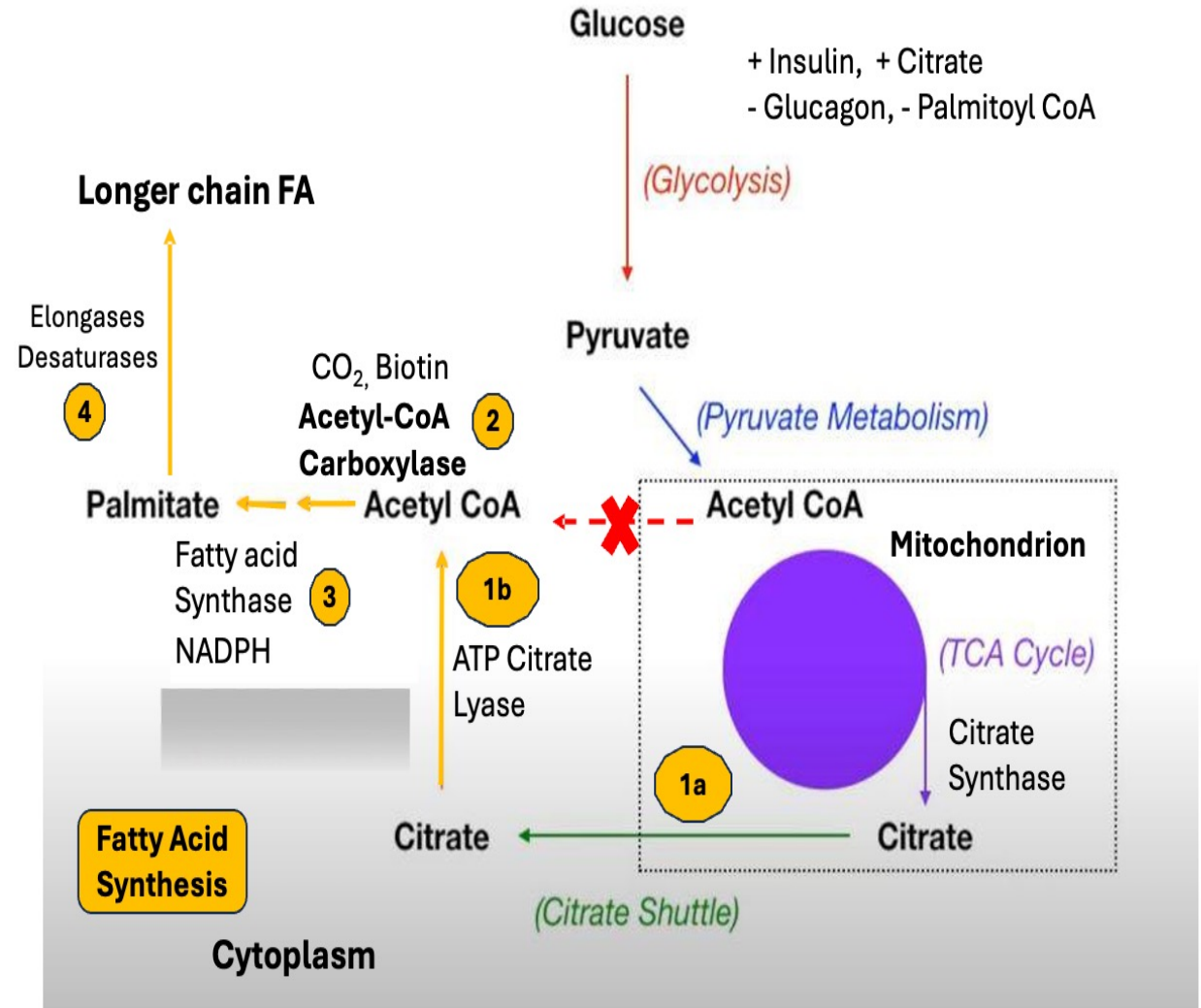


(a) The general structure of a fatty acid (22). (b) The chemical structure of α -linolenic acid (ALA), 18:3n-3. ALA has 18 carbon atoms (C) and 3 double bonds, the first of which is located 3 carbon atoms from the terminal methyl group (omega end).



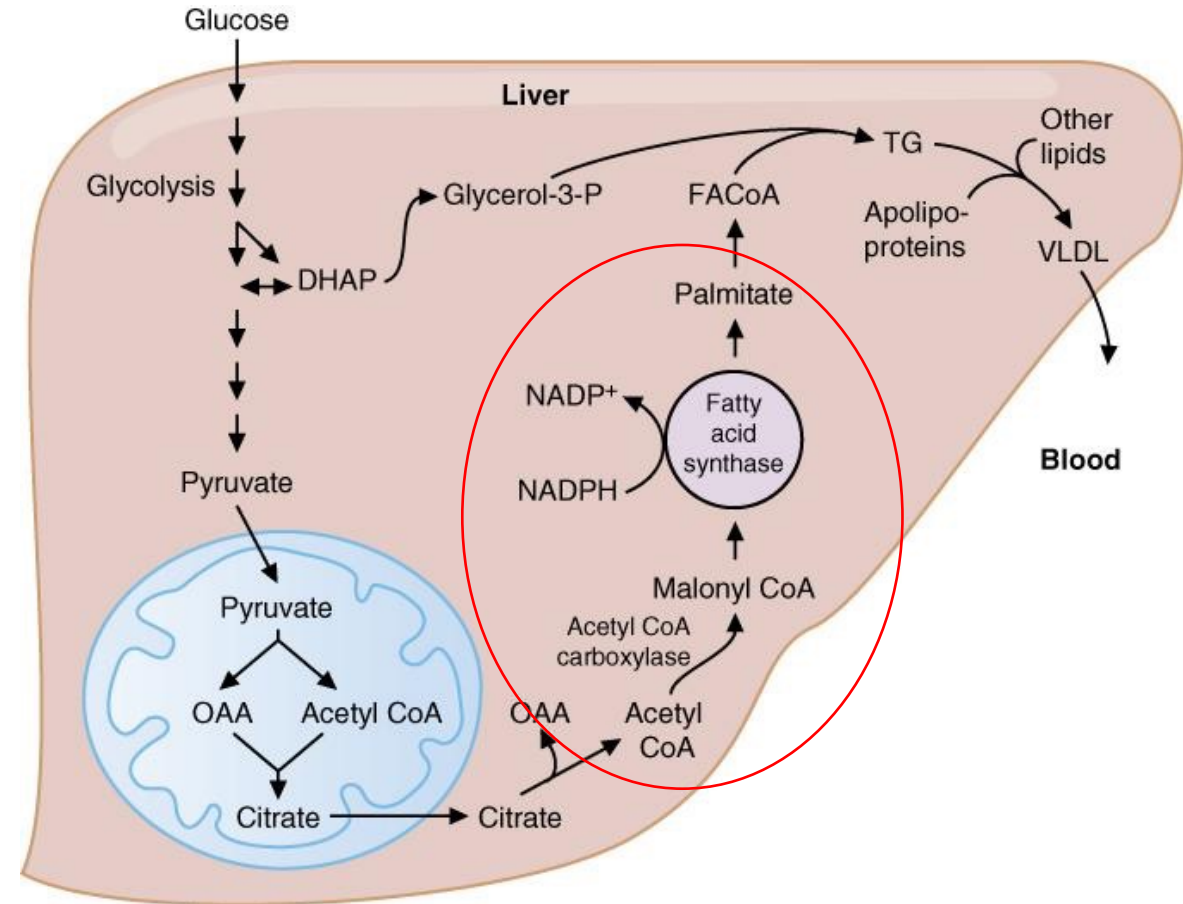
Overview of Fatty Acid Synthesis

- **Excess Glucose** → Pyruvate → **Acetyl-CoA**
- **Acetyl-CoA EXITS mitochondria AS CITRATE** which is then converted back to **Acetyl-CoA** by **ATP Citrate Lyase**
- **Acetyl-CoA → Malonyl-CoA** by **Acetyl-CoA Carboxylase (ACC)**
- **Fatty Acid Synthase (FAS)** condenses one **Acetyl-CoA** & 7 **Malonyl-CoA** into **Palmitate** which then can be modified into various Fatty acids forms



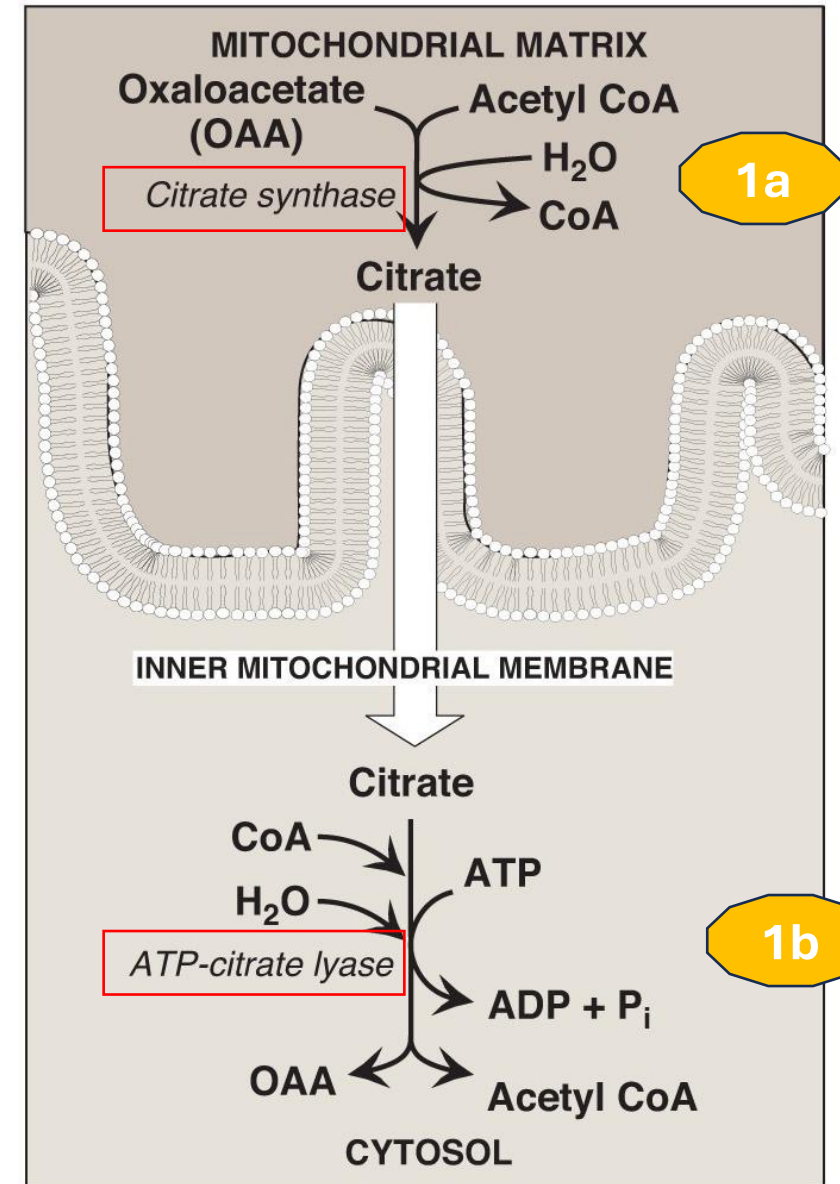
Mechanism of *de novo* Synthesis of Fatty Acids

1. Production of **cytosolic Acetyl-CoA**.
2. Formation of **Malonyl CoA** by **ACC**
ACC requires ATP & CO₂ & Biotin
3. Synthesis of **Palmitate (16C Sat-FA)**
by **FA Synthase (FAS)**. Requires **NADPH**
4. **Modification of Palmitate**
 - Elongation of palmitate by **Elongases**
 - Desaturation by **Desaturases**



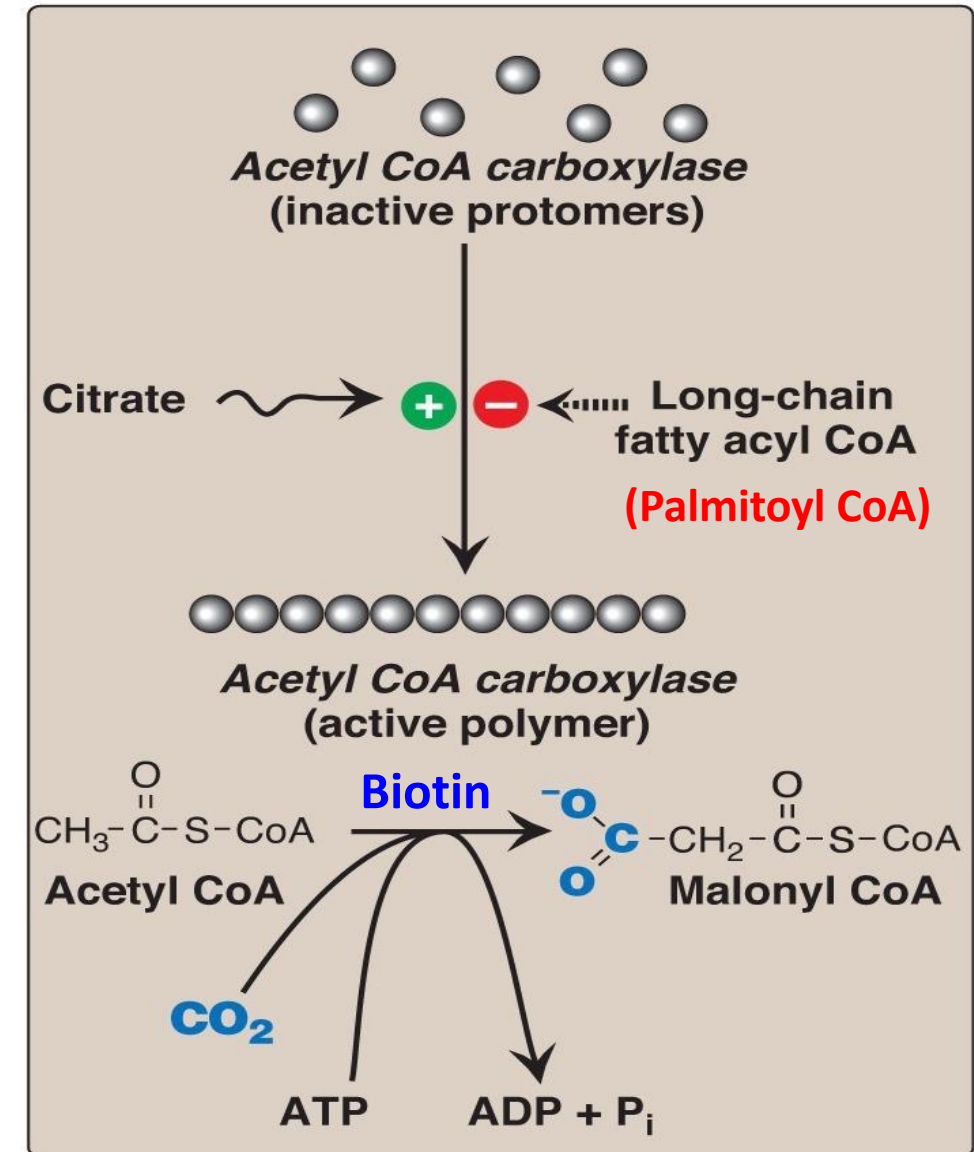
Production of Cytosolic Acetyl CoA

- **Acetyl CoA formed in mitochondria cannot cross mitochondrial membrane into cytosol.**
- **Acetyl-CoA converted to Citrate**
- **Citrate leaves into the cytosol**
- **ATP-Citrate Lyase cleaves Citrate in cytosol to form Acetyl-CoA & OAA**



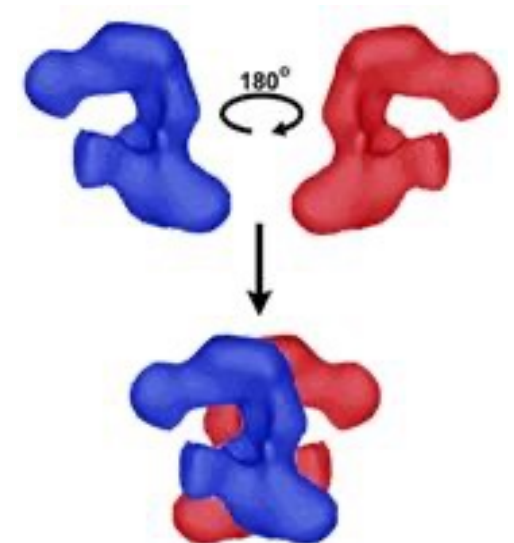
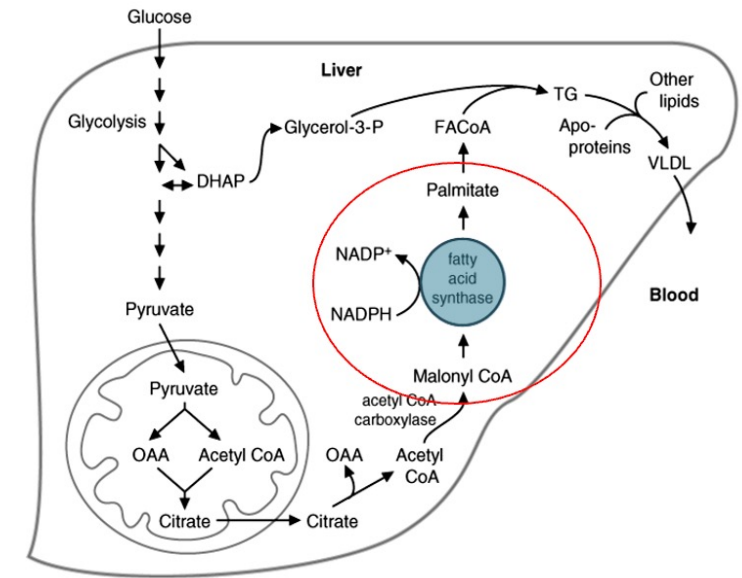
Acetyl CoA carboxylase (ACC): Acetyl CoA $\rightarrow$ Malonyl CoA

- **Acetyl-CoA $\rightarrow$ Malonyl-CoA is the RLS**
- **ACC = Regulated** step of FA Synthesis
- **ACC = ABC** enzyme
 - **Uses ATP**
 - Requires **Biotin** (Vitamin **B7**)
 - **Adds CO₂** to Acetyl-CoA



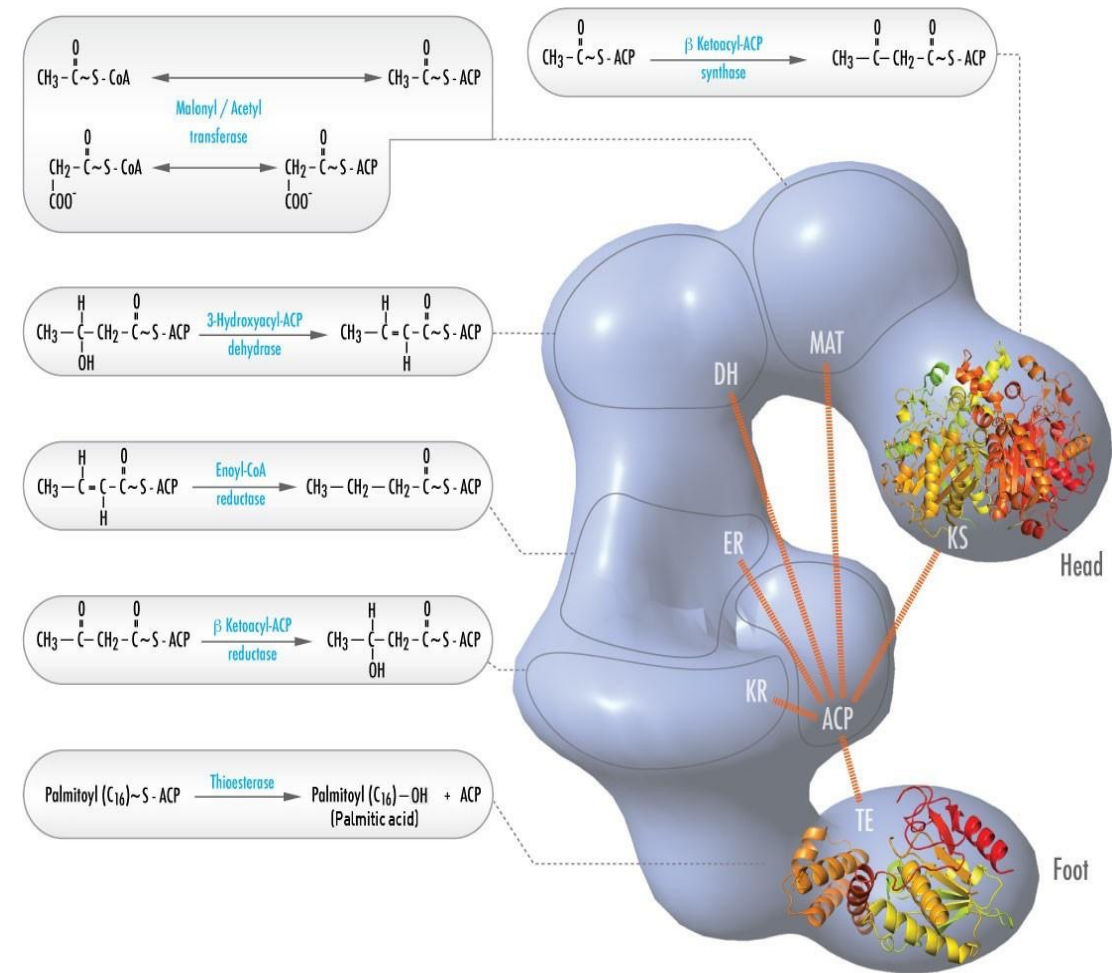
Fatty Acid Synthase complex (FAS)

- **FAS** is a **multifunctional** enzyme that **catalyzes a series of rxns** whereby **2 carbon units from Malonyl-CoA** are **linked together**, ultimately to form **Palmitoyl-CoA**
- **Acetyl-CoA & 7 Malonyl-CoA** are **condensed into a Palmitate (16C)**
- The **enzyme complex is a dimer**
- Each monomer has **7 different enzyme activities** & contains an **acyl-carrier domain (ACP)** & a contains a **temporary holding domain** with a **Cysteine** residue
- **Insulin UPREGULATES** the amount of enzyme expressed



Fatty Acid Synthase Mechanism

- Acetyl-CoA & Malonyl-CoA
CONDENSATION
- Repeating cycle of
Reduction requires NADPH
Reduction & Dehydration
ADDS 2-carbons in each cycle to
the FA chain (Elongation)
- End product = **Palmitate (16C FA)** released from the enzyme.

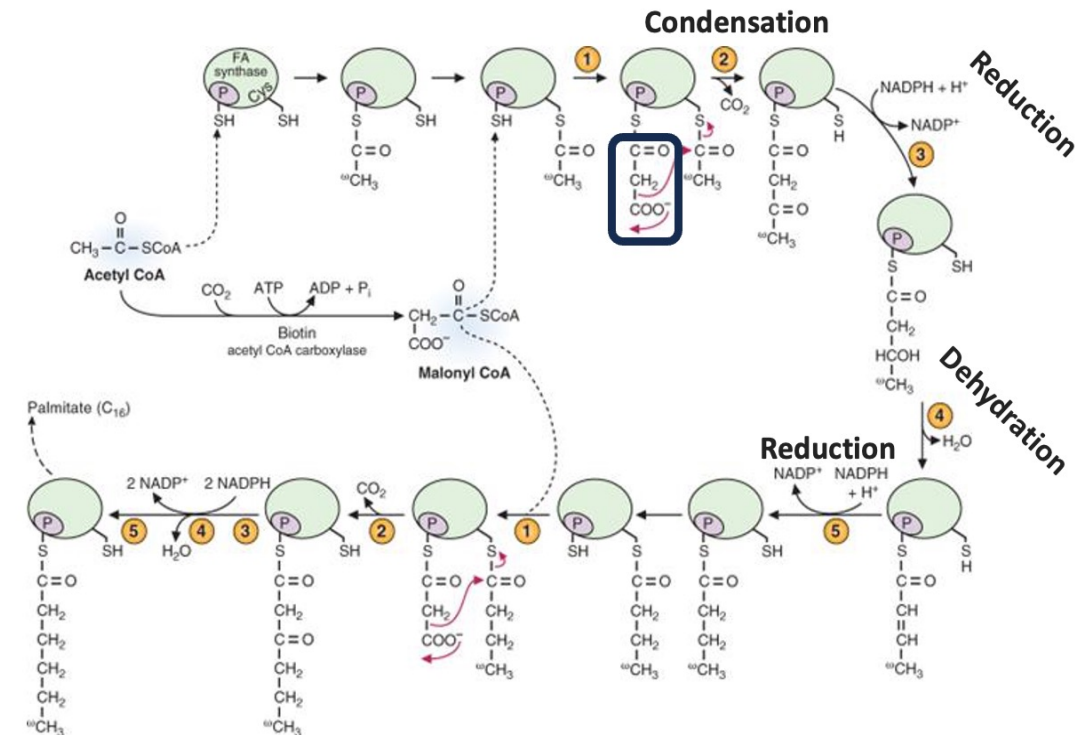


Synthesis of Palmitate in FAS Complex

- Acetyl- CoA is attached to FAS
- Malonyl- CoA enters and provides the 2-C units that are added to the growing Fatty-acyl chain

Following steps are repeated until Palmitate (16C) is produced

1. Addition of Malonyl-group to the Phosphopantotheinyl residue
2. Condensation of Malonyl + Fatty-acyl groups
3. Reduction of β -ketoacyl group (NADPH required)
4. Dehydration
5. Reduction of the double-bond (NADPH required)

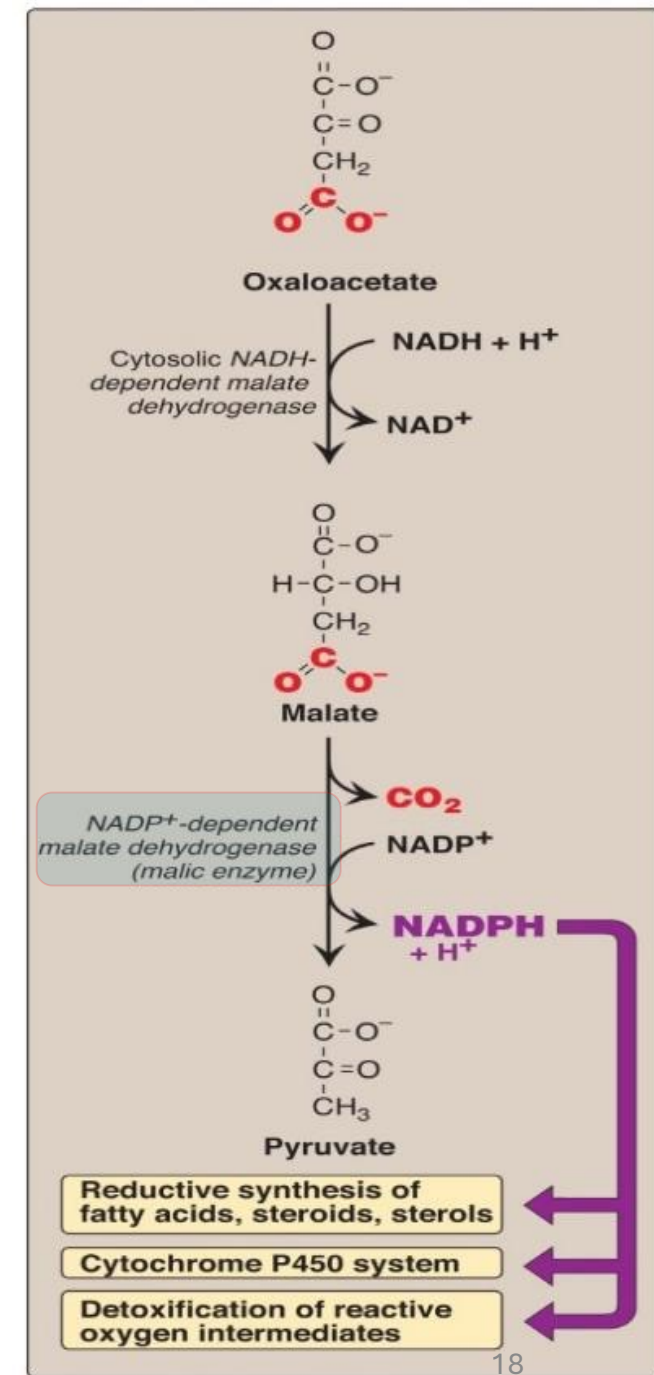


Sources of NADPH for Fatty Acid Synthesis

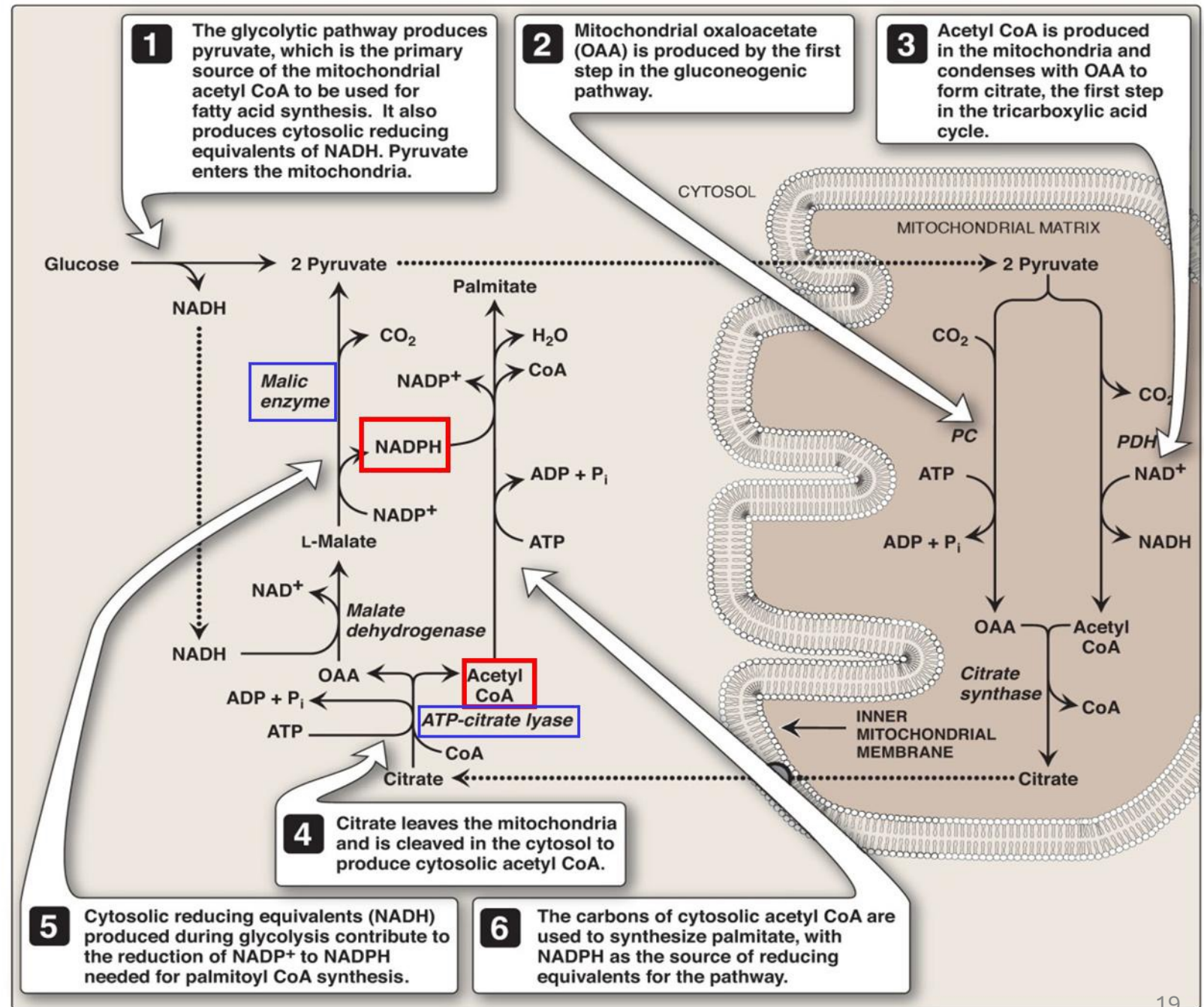
The **NADPH** required to provide **reducing power** for **FA Synthesis** is produced by =

1. PPP (major source) (G6PDH, cytosol)

2. Malic enzyme (NADP-dependent Malate Dehydrogenase) (Malate → Pyruvate)



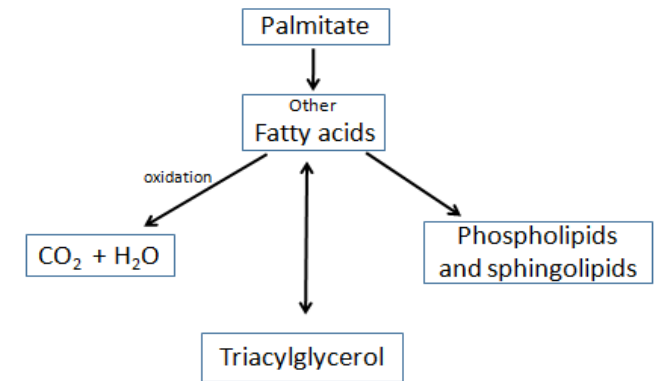
Transport of acetyl units and (NADPH) reducing equivalents used in FA synthesis



Modification of Dietary & Endogenous Fatty Acids

Palmitate produced is **modified to give rise to the other Fatty acids**

Fatty acids from diet may also be **modified**

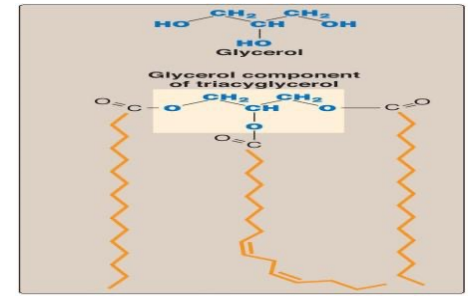


Modifications include

- Chain **Elongation** (occurs in **Endoplasmic Reticulum**) to produce longer fatty acids:
- Elongation by enzymes known as **Elongases** (different from FAS)
- **Brain makes Very-Long-Chain FAs (VLCFAs 22C-24C) for synthesis of brain lipids**
- **Desaturation** (occurs in **Endoplasmic Reticulum**), producing unsaturated fatty acids: Fatty Acyl-CoA Desaturases **introduce double-bonds @ positions 5, 6 or 9 in humans.**

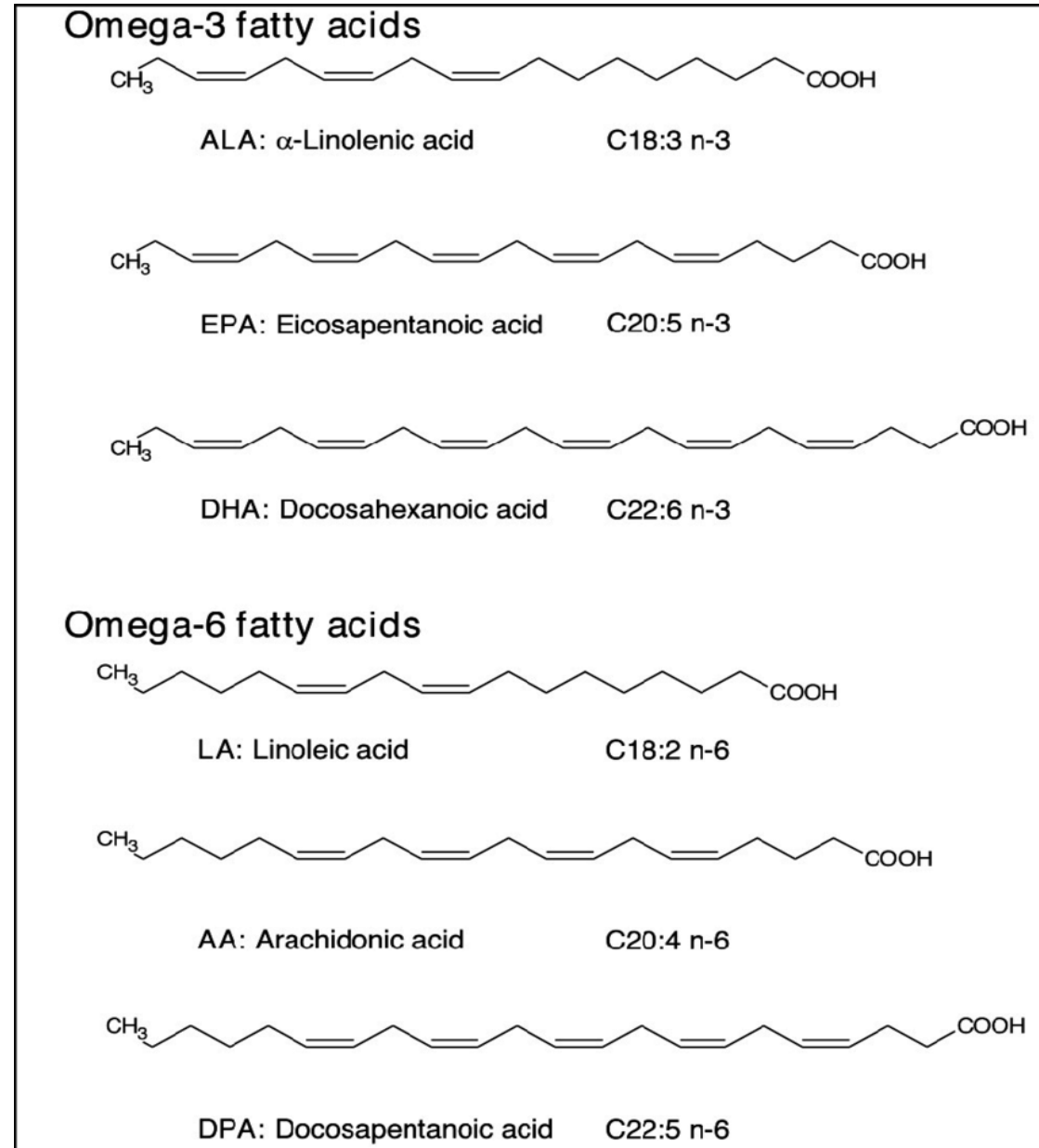
Essential Fatty Acids

- **Mammals lack enzymes that can Desaturate beyond the $\Delta 9$ position in the hydrocarbon chain**
- **Plants, including algae, can desaturate beyond the $\Delta 9$ position (ie. $\Delta 12$, $\Delta 15$ desaturases)**
- **Dietary Essential FAs (CANNOT be synthesized in humans and must be provided in diet)**
- **Linoleate (omega-6)**
- **α -Linolenate (omega-3)**



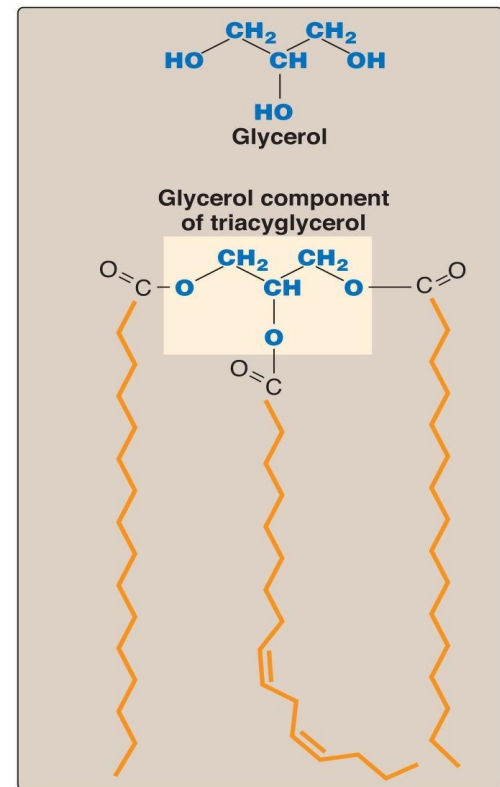
Essential Fatty Acids = Linolenic acid & Linoleic acid

- ω 6 family CANNOT be converted to a FA of ω 3 family.
- **Linolenic acid (ω 3) $\rightarrow$ EPA $\rightarrow$ DHA**
- **DHA** in **brain phospholipids** = provides high **fluidity**
- **DHA** is important for **brain functions** and the **vision**
- **Linoleic acid (ω 6) $\rightarrow$ Arachidonic acid**
- **Arachidonic acid** used to **form Eicosanoids** (prostaglandins, thromboxane, leukotrienes)



TAGs

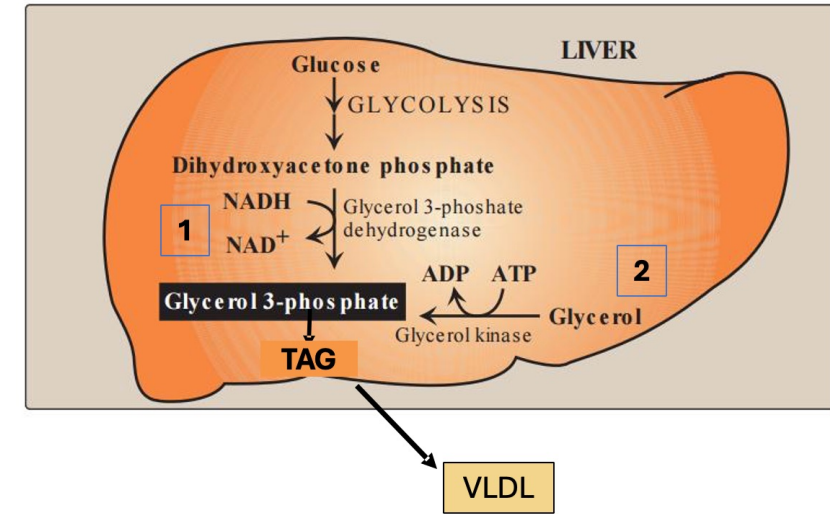
- **FA are stored as TAGs (triglycerides)**
- **Glycerol + 3 FAs**
- **Glycerol 3-P produced in = Liver & Adipose tissue.**
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- **TAG synthesis = Liver & Adipose & Intestinal Mucosa**
- **Secretion of VLDL from the Liver is an important mechanism to prevent the accumulation of TAG in the Liver**



TAG Synthesis in Liver

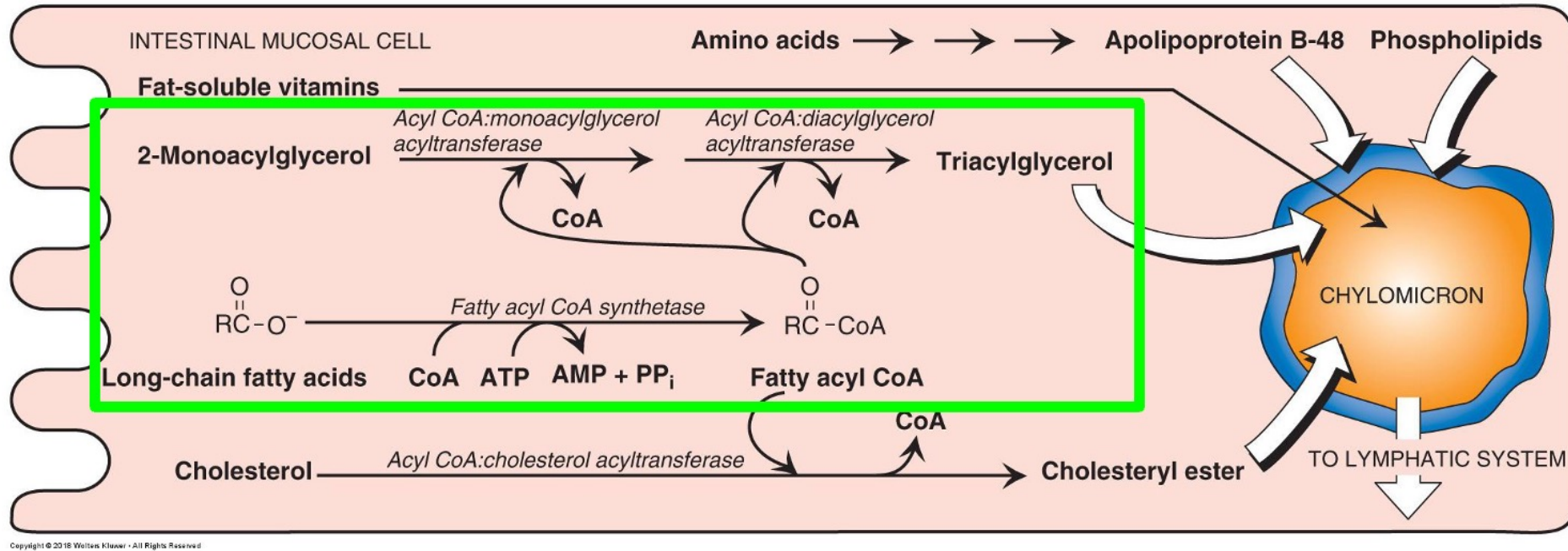
Source of Glycerol 3-Phosphate

- **DHAP** (from Glycolysis) in **Liver** and **Adipose** tissue
- **Glycerol + Glycerol Kinase = Glycerol 3-Phosphate**
- **Glycerol Kinase** present in **Liver**; absent in adipose tissue



- **3 FAs added sequentially** to form Triacylglycerol **TAG**
- **TAG** produced in the **Liver** is packaged with apolipoproteins to form **VLDL**
- **VLDL** secreted **into blood** and transported to **Adipose & Muscle** where they **deliver FAs** to these tissues (Lipoprotein-Lipase)
- **Excessive storage of TAG in Liver** = results in **Fatty Liver disease**

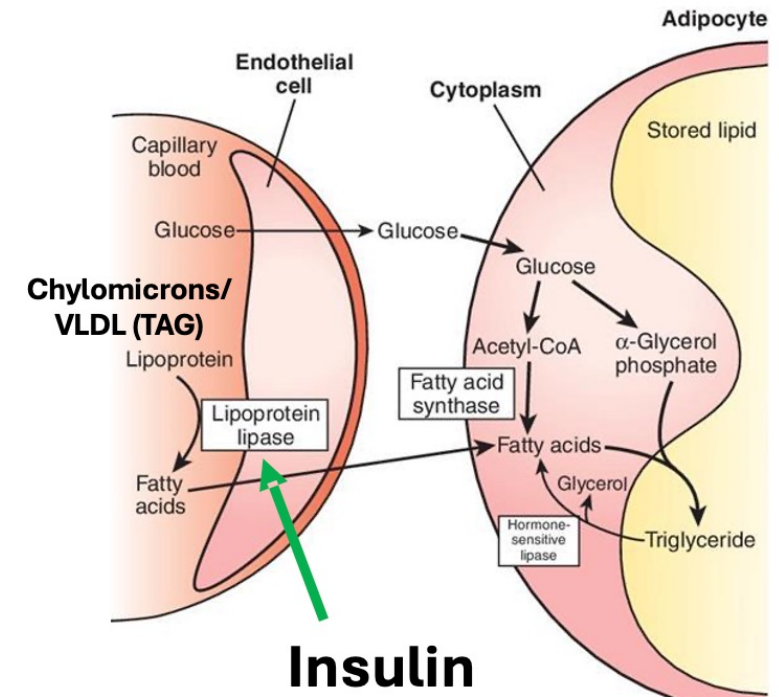
Synthesis of TAGs in Intestinal mucosal cells (MAG Pathway)



- **Monoacylglycerol (MAG) = End-product of Lipid digestion**
- **2 FAs added to MAG to form TAG**
- **TAG released as Chylomicron → FA delivery to Adipose & Muscle**

Delivery of FAs to Adipocytes – *Lipoprotein-Lipase*

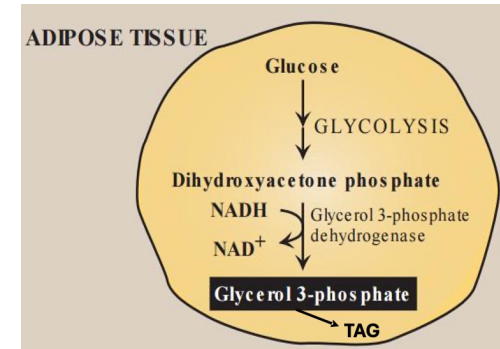
- **TAGs in Chylomicrons & VLDL are hydrolyzed to free FA by Endothelial Lipoprotein-Lipase, liberated FAs enter Adipocyte where they will be stored as TAGs**
- **Lipoprotein-Lipase is activated by Insulin (i.e. Insulin facilitates TAG storage**



TAG Synthesis in Adipose tissue

Source of Glycerol 3-Phosphate in Adipose tissue

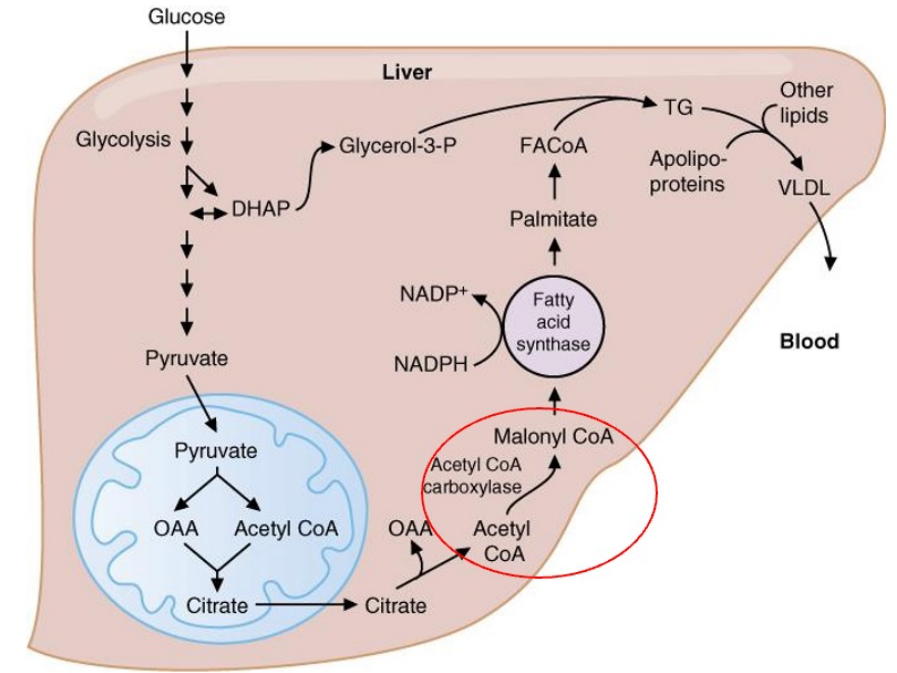
- **DHAP** from Glycolysis (in both Liver & Adipose tissue)
- **DHAP** ---Glycerol 3-Phosphate Dehydrogenase-----> Glycerol-3-Phosphate
- **Adipocytes DO NOT** have Glycerol Kinase to preform Glycerol → Glycerol-3-Phosphate



-
- **3 FAs** are then **added** sequentially to **form TAG**
 - **TAG STORED** in **WHITE FAT** serve as the **major energy reserve** of the body
 - This energy store when **in excess results in obesity**
 - **FAs** are **released** by **Lipolysis** during **fasting / exercise /stress**
 - **TAG** in **Brown Adipocytes** serves as source of heat through **non-shivering thermogenesis**

Regulation of *de novo* Synthesis of Fatty Acids

- **Acetyl-CoA Carboxylase (ACC)** is the **RLS** enzyme in **FA Synthesis**
- **Acetyl-CoA Carboxylase (ACC)** is the **first committed step** of **FA Synthesis**



Regulation of *de novo* synthesis of FAs

Regulation of Acetyl-CoA Carboxylase (ACC)

Substrate Availability

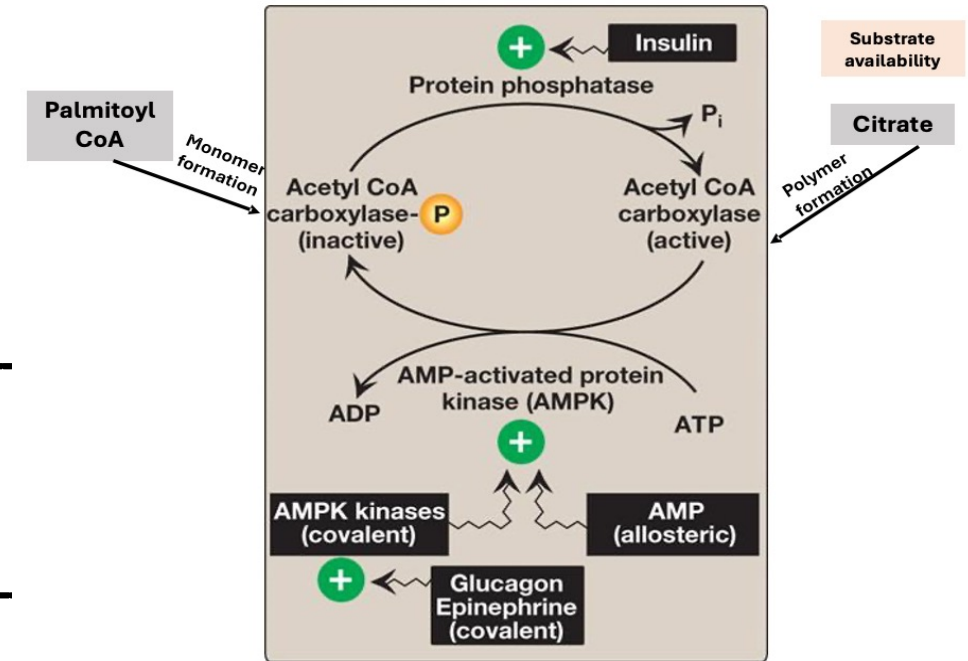
- CYTOSOLIC Acetyl-CoA

Allosteric

- Activated by Citrate
- Inhibited by Palmitoyl-CoA & AMP (Monomer is inactive, Polymer still active)

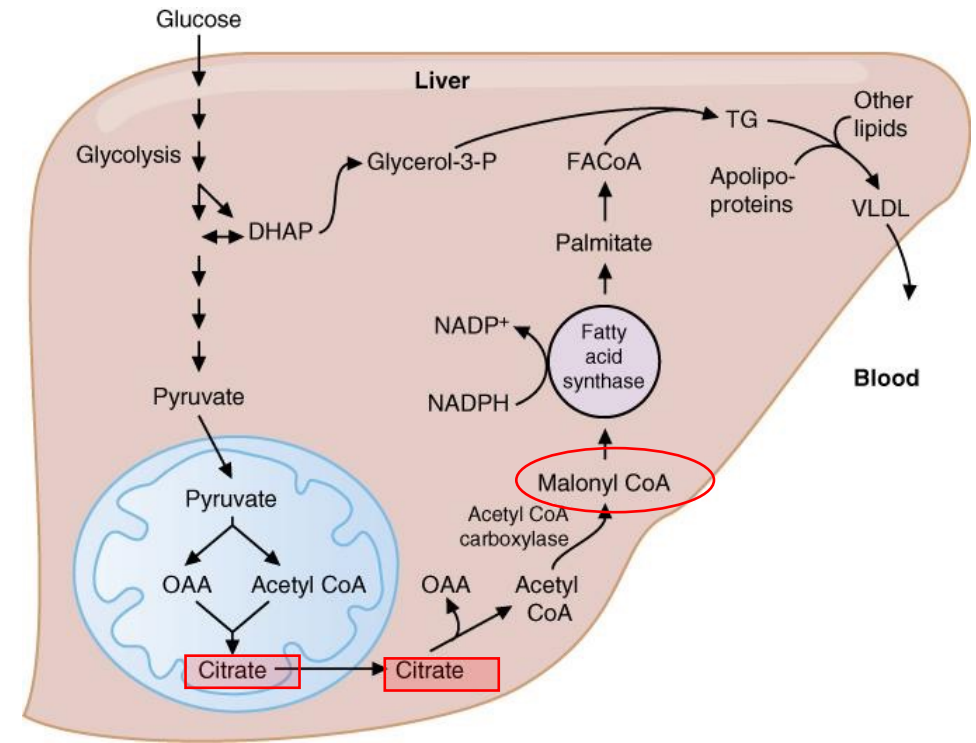
Covalent Modification

- Activated by Dephosphorylation via Insulin
- Inhibited by Phosphorylation via Glucagon
- **Active** in fed state high blood glucose **Insulin**
- **Inhibited** in **fasted** state low blood glucose, **Glucagon**



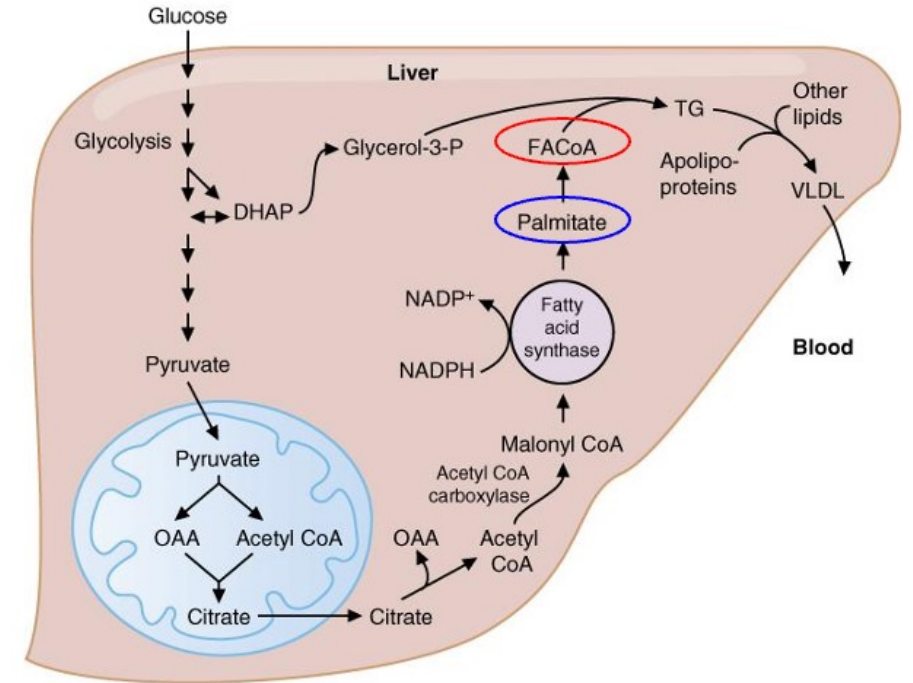
Acetyl CoA Carboxylase (ACC)

- **Citrate** is found in **CYTOSOL** when **mitochondrial citrate** levels are **very high**
- **Cytosolic** citrate is used for *de-novo* synthesis of **FA & Cholesterol Synthesis**
- **Cytosolic Citrate FEED-FORWARD** activates **ACC** and leads to polymer formation.



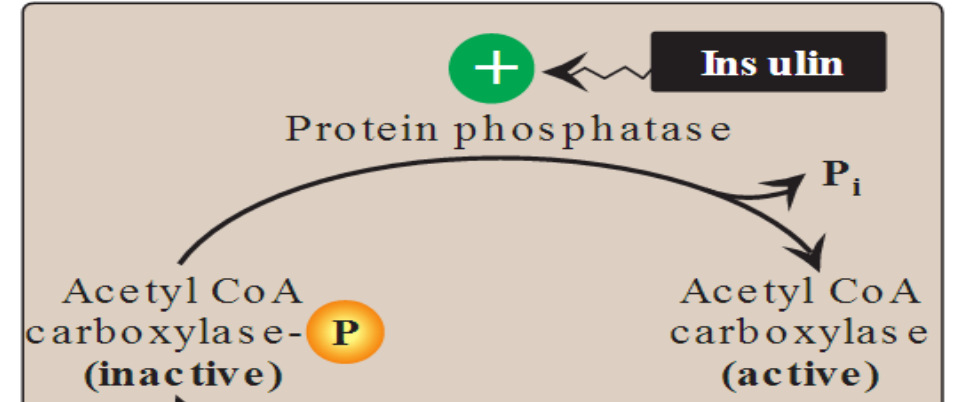
Acetyl CoA Carboxylase (ACC)

- **Accumulation of Palmitoyl-CoA**, Long-chain Fatty Acyl-CoA (*de novo* FA Synthesis end-product) allosterically **inhibits ACC**
- **Favors Monomer form = ACC Less active**
- **Palmitoyl-CoA induces FEED-BACK inhibition on ACC**



Activation of ACC by Covalent Modification

- ACC active when Dephosphorylated
- Insulin favors *de-novo* synthesis of FAs
- Insulin = activates Phosphoprotein Phosphatase which removes the inhibitory-regulatory phosphate on ACC = allowing it to become active
- Insulin upregulates ACC

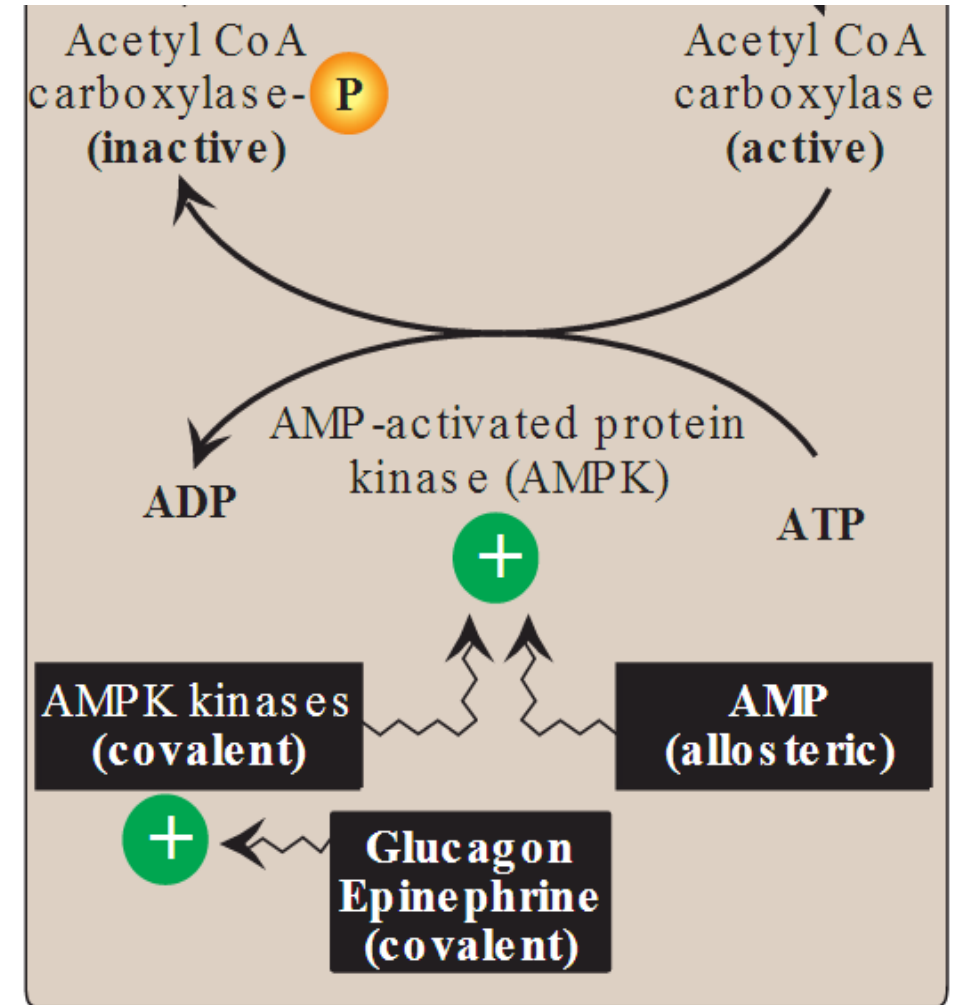


Inhibition of ACC

- **ACC inhibited when Phosphorylated**
- **Inhibitory Phosphorylation by AMPK**

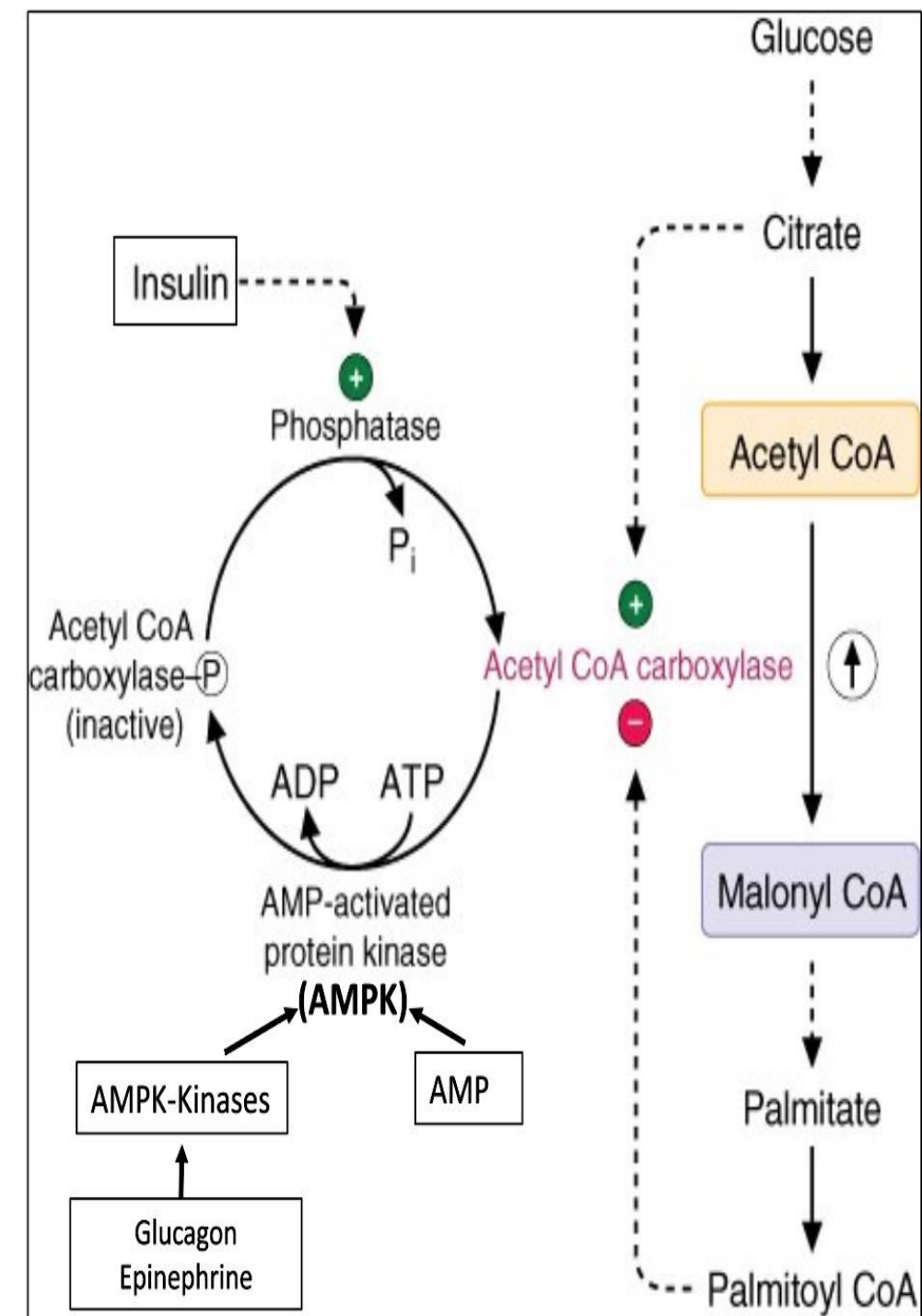
Activation of AMPK

- **Allosteric AMP**
- **Glucagon signal**
- **Epinephrine signal**
- **Norepinephrine signal**



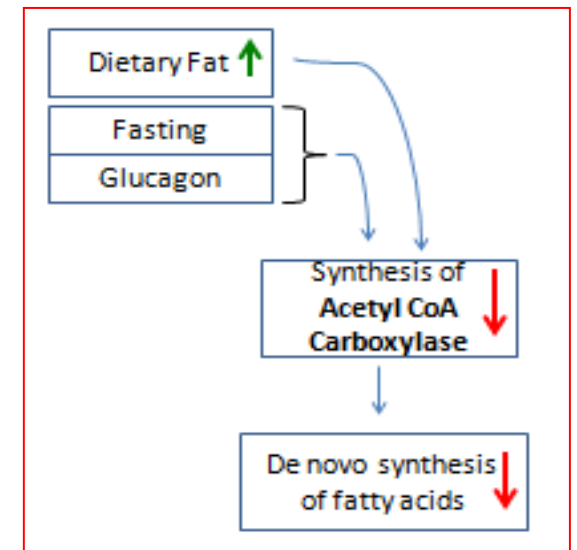
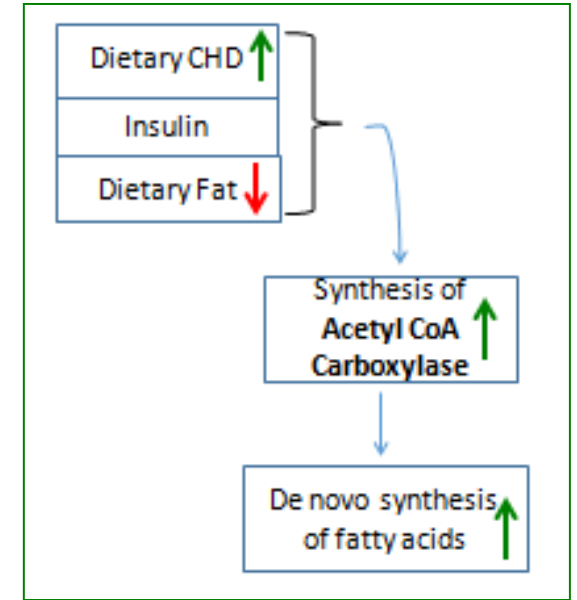
SHORT-TERM Regulation of ACC

- **Substrate = Cytosolic Citrate**
- **Allosteric = activated** by **Citrate** (polymer formation) and **inhibited** by **Palmitoyl-CoA** (monomer formation).
- **Covalent = activated** by **Dephosphorylation** (Insulin) and **inhibited** by **Phosphorylation** (Glucagon, Epinephrine)
- Enzyme **induction by Insulin**



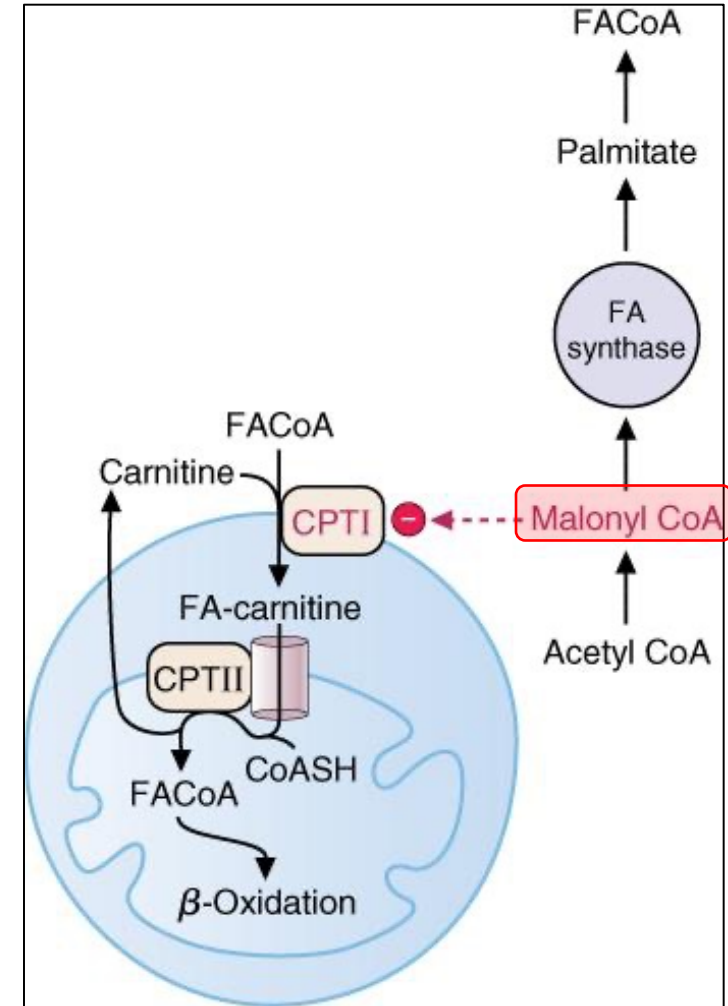
LONG-TERM Regulation of ACC

- **High Carbohydrate and Fat-Free diets and Insulin lead to increased synthesis of ACC (up-regulation, induction) resulting in increased FA Synthesis**
- **High-fat diets, Fasting and Glucagon lead to decreased synthesis (down-regulation, repression) resulting in the outcome of decreased FA Synthesis**



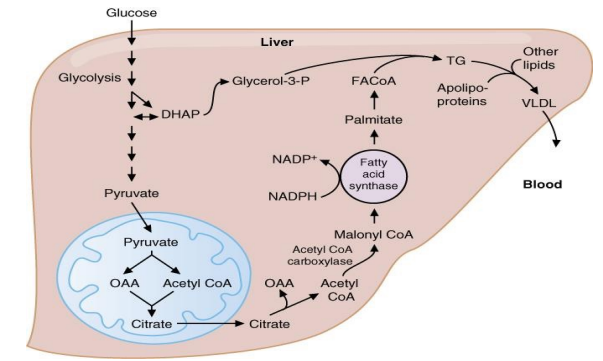
Malonyl-CoA Inhibition of Carnitine-Palmitoyl Transferase I (CPT-I)

- **Malonyl-CoA** formed in the **Cytosol** during **FA synthesis INHIBITS** the **CPT-I** channel which is involved in the **transports of Long-chain FAs into the Mitochondria** for **β -oxidation**
- **Ensures** that **during FA synthesis**, that its opposing **β -oxidation** is **inhibited** (reciprocal regulation)
- **Newly synthesized FA** remain in the **cytosol** and are **not immediately oxidized** in the **fed state**



Summary

- **FAs are synthesized in Liver, primarily from Glucose**
- **Glucose** converted to **Pyruvate** (Glycolysis) → **enters mitochondria** and **forms Acetyl-CoA**; Acetyl CoA and **OAA** → **Citrate**
- **Citrate** transported to **cytosol**, forms **Acetyl-CoA** (by **ATP-citrate lyase**) = **source of carbons for FA Synthesis**
- Enzymes: **ACC (key regulatory step)** & **FAS**, produce **Palmitic acid (16C, saturated)**
- **Insulin** (well-fed state) **activates** fatty acid **synthesis** pathway
- **Palmitoyl-CoA** can be **elongated** or **desaturated** (adding double bonds) **by enzymes in Endoplasmic Reticulum**
- **FAs used to produce TAGs** for **energy storage in Adipocytes**, and **Glycerophospholipids & Sphingolipids** for structural components of cell **membranes**



Fatty acid Synthesis - Summary

	SYNTHESIS
Greatest flux through pathway	After carbohydrate-rich meal
Hormonal state favoring pathway	High insulin/glucagon ratio
Major tissue site	Primarily liver
Subcellular location	Primarily cytosol
Carriers of acyl/acetyl groups between mitochondria and cytosol	Citrate (mitochondria to cytosol)
Phosphopantetheine-containing active carriers	Acyl carrier protein domain, coenzyme A
Oxidation/reduction coenzymers	NADPH (reduction)
Two-carbon donor/product	Malonyl CoA: donor of one acetyl group
Activator	Citrate
Inhibitor	Long-chain fatty acyl CoA (inhibits <i>acetyl CoA carboxylase</i>)
Product of pathway	Palmitate
Repetitive four-step process	Condensation, reduction, dehydration, reduction